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Enhanced stroke risk prediction in hypertensive patients through deep learning integration of imaging and clinical data

Hui Li¹, Tianyu Zhang², Guochao Han¹, Zonghui Huang², Huiyu Xiao², Yunzhe Ni², Bo Liu³, Wennan Lin⁴ and Yuan Lin^{1*}

Abstract

Background Stroke is one of the leading causes of death and disability worldwide, with a significantly elevated incidence among individuals with hypertension. Conventional risk assessment methods primarily rely on a limited set of clinical parameters and often exclude imaging-derived structural features, resulting in suboptimal predictive accuracy.

Objective This study aimed to develop a deep learning-based multimodal stroke risk prediction model by integrating carotid ultrasound imaging with multidimensional clinical data to enable precise identification of high-risk individuals among hypertensive patients.

Methods A total of 2,176 carotid artery ultrasound images from 1,088 hypertensive patients were collected. ResNet50 was employed to automatically segment the carotid intima-media and extract key structural features. These imaging features, along with clinical variables such as age, blood pressure, and smoking history, were fused using a Vision Transformer (ViT) and fed into a Radial Basis Probabilistic Neural Network (RBPNN) for risk stratification. The model's performance was systematically evaluated using metrics including AUC, Dice coefficient, IoU, and Precision-Recall curves.

Results The proposed multimodal fusion model achieved outstanding performance on the test set, with an AUC of 0.97, a Dice coefficient of 0.90, and an IoU of 0.80. Ablation studies demonstrated that the inclusion of ViT and RBPNN modules significantly enhanced predictive accuracy. Subgroup analysis further confirmed the model's robust performance in high-risk populations, such as those with diabetes or smoking history.

Conclusion The deep learning-based multimodal fusion model effectively integrates carotid ultrasound imaging and clinical features, significantly improving the accuracy of stroke risk prediction in hypertensive patients. The model

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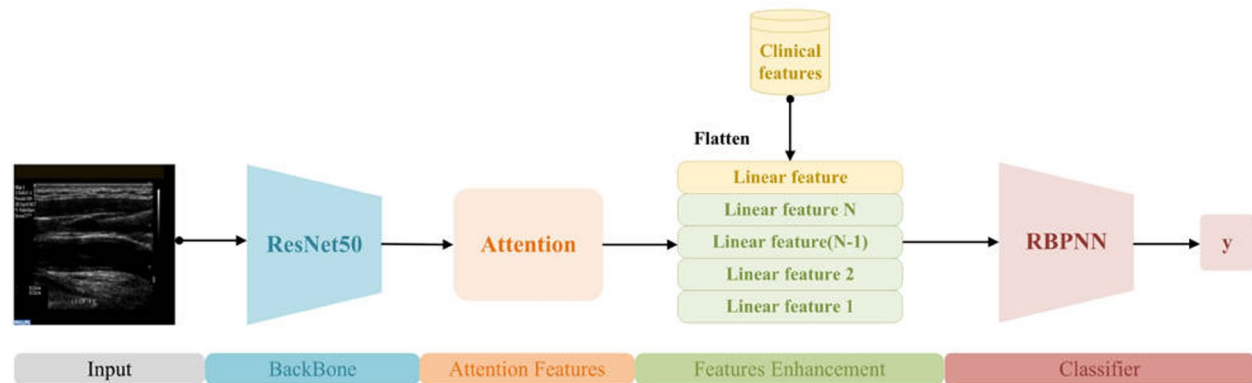


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demonstrates strong generalizability and clinical application potential, offering a valuable tool for early screening and personalized intervention planning for stroke prevention.

Clinical trial number Not applicable.

Graphical Abstract



Keywords Stroke prediction, Deep learning, Carotid artery ultrasound, Multimodal data fusion, ResNet50, Vision transformer

Background

Stroke is one of the most common causes of death and disability worldwide, posing a significant threat to human health. Data from the World Health Organization (WHO) indicate that stroke is the second most common cause of death and the primary contributor to persistent disability globally [1]. In China, both the incidence and mortality rates of stroke have been steadily increasing, presenting a growing public health challenge [2]. The incidence of stroke is notably higher among hypertensive patients compared to the general population, and the risk increases significantly with age [3, 4]. Hypertension is one of the primary risk factors for stroke, as blood pressure fluctuations contribute to endothelial injury and accelerate the development of arteriosclerosis, thereby directly exacerbating the risk of stroke [5, 6]. Identification of high-risk individuals among hypertensive patients, combined with timely and effective interventions, is crucial for reducing stroke incidence and mortality.

Conventional methods for predicting stroke risk primarily rely on individual clinical parameters, including blood pressure, blood glucose, and lipid levels, or imaging results, such as computed tomography (CT) or magnetic resonance imaging (MRI) [7]. However, these single-source prediction methods have significant limitations, as their performance is frequently compromised by data noise and incompleteness, leading to decreased predictive accuracy. Advances in medical imaging and data processing have facilitated the integration of multimodal data, offering a promising strategy to enhance the precision of stroke risk prediction [8]. Carotid ultrasound, as a

non-invasive, accessible, and efficient imaging technique, provides direct visualization of vascular endothelial damage and the extent of atherosclerosis, and has been widely adopted in stroke risk assessment [9, 10]. Additionally, clinical variables such as age, smoking status, and body mass index (BMI) contribute valuable physiological information that complements imaging data. The integration of multimodal inputs allows for a more comprehensive evaluation of the patient's health status, thereby improving the accuracy and reliability of stroke risk prediction.

Deep Learning (DL) techniques, particularly Convolutional Neural Networks (CNNs) and self-attention mechanisms such as Transformers, have led to revolutionary advancements in medical image analysis in recent years [11–13]. These models automatically learn features from large-scale data and efficiently perform feature extraction and classification, significantly enhancing the automation and accuracy of image analysis. Residual Neural Network50 (ResNet50), a deep CNN, introduces residual connections to address gradient vanishing and explosion issues during training, demonstrating robust feature extraction performance [14]. In contrast, the Vision Transformer (ViT) model divides images into smaller patches and uses the self-attention mechanism for feature extraction, allowing more effective capture of spatial relationships and global contextual information. Combining these two DL models improves the segmentation and recognition accuracy of medical images while improving generalization performance and reducing overfitting risk [15]. In stroke risk prediction, DL techniques extract critical features, such as arterial intima-media thickness

and vessel stenosis, and integrate these features with clinical variables to produce more precise and individualized predictive outcomes.

In recent years, growing interest has emerged in applying artificial intelligence (AI) technologies to improve the integration and classification of omics data, particularly for identifying disease subtypes from multi-omics datasets. Several studies have explored AI-based approaches in this domain. For instance, cat swarm optimization (CSO) combined with support vector machines (SVM) has been employed to classify and cluster cancer subtypes (*Journal of Soft Computing and Data Mining*, 2023 [16]). Other investigations have integrated quantum-enhanced CSO with multi-omics clustering algorithms to improve subtype discrimination in oncology (*Iraqi Journal for Computer Science and Mathematics*, 2023 [17]). A recent review provided a comprehensive summary of progress and ongoing challenges in AI-driven omics data analysis, emphasizing limitations in feature interpretability, cross-modal data integration, and model generalization in scenarios involving limited sample sizes (*International Journal of Mathematics, Statistics, and Computer Science*, 2023 [18]).

However, the majority of previous studies have focused on static omics data, including genomics and transcriptomics, primarily for disease classification and subtype identification. Research targeting personalized risk prediction in chronic progressive diseases such as stroke, especially integrating structural imaging with dynamic clinical variables, remains limited. This study addresses this research gap by proposing a deep multimodal learning framework that combines semantic image segmentation, global attention modeling, and nonlinear fusion-based classification to enable practical and forward-looking stroke risk prediction in hypertensive patients.

Although recent studies have explored the use of multimodal data in stroke risk prediction, notable limitations persist. Researchers such as M. Bendechache [19] and S. Anari [20] have contributed meaningful progress in the integration of structured clinical data and the development of neural network models, offering effective paradigms for disease risk assessment. However, these approaches have largely emphasized shallow integration of phenotypic data or conventional scoring systems, with limited attention to deep modeling strategies based on structural features extracted from carotid ultrasound imaging.

First, many existing studies still rely on single-source data without systematically integrating imaging information with multidimensional clinical variables, which restricts their ability to accurately characterize real-world clinical conditions. Second, most predictive models are based on traditional machine learning algorithms

or shallow network architectures, restricting the ability to extract fine-grained imaging features such as intima-media thickness and vessel boundary morphology. As a result, these models offer insufficient support for detailed and accurate risk assessment. Lastly, individual variability is often overlooked in model design, resulting in limited generalizability across different populations and clinical settings.

To address these challenges, this study proposes a deep learning-based multimodal fusion model that employs ResNet50 for automatic segmentation of the common carotid artery intima in ultrasound images, utilizes Vision Transformer (ViT) for global feature modeling, and integrates a Radial Basis Probabilistic Neural Network (RBPNN) to fuse imaging and multidimensional clinical features. The proposed framework enhances the sensitivity to structural imaging cues and improves the nonlinear representation of clinical variables, achieving an optimal balance between prediction accuracy and generalizability. It offers a more advanced and practical solution for individualized stroke risk assessment and the early identification of high-risk individuals.

The primary objective of this study is to develop a stroke risk prediction model for high-risk patients by integrating DL techniques with carotid ultrasound images and multimodal clinical data. A total of 2,176 carotid ultrasound images were collected from 1,088 hypertensive patients, along with corresponding clinical data (e.g., age, smoking history, blood pressure). The combined application of ResNet50 and ViT enabled automated and precise analysis of both imaging and clinical data to support accurate risk prediction. The clinical significance of this study lies in the development of an efficient and stable early screening tool for hypertensive patients, enabling timely identification of individuals at elevated risk for stroke. The predictive framework supports the implementation of personalized interventions and treatment strategies based on individual risk profiles. By providing a scientific foundation for tailored prevention and management of hypertension, the model contributes to reducing both the incidence and disability associated with stroke, thereby enhancing long-term health outcomes and quality of life in affected populations.

The main contributions of this study are summarized as follows

This study is the first to propose a deep learning framework that integrates carotid ultrasound imaging with multidimensional clinical data for precise stroke risk stratification in hypertensive patients.

A novel semantic modeling approach combining ResNet50 for segmentation and Vision Transformer (ViT) for global image representation is introduced,

which significantly improves the extraction and characterization of carotid intima-media features.

A Radial Basis Probabilistic Neural Network (RBPNN) is incorporated to enable nonlinear high-dimensional decision fusion of imaging and clinical features, leading to enhanced predictive performance.

Extensive ablation studies and subgroup analyses are conducted to systematically evaluate the contributions of each model component and assess robustness across high-risk patient subgroups.

The computational complexity and clinical deployment feasibility of the proposed model are assessed, demonstrating its strong generalizability and practical application potential.

The remainder of this paper is organized as follows

Section “[Materials and methods](#)” describes the dataset sources and preprocessing procedures. Section “[Results](#)” details the architecture of the proposed multimodal fusion model. Section “[Discussion](#)” presents the experimental design and analysis of results. Section “[Conclusions](#)” discusses the findings and clinical implications. Section 6 concludes the paper and outlines future research directions.

Materials and methods

Source and collection of carotid ultrasound data

This study aims to construct a DL model that integrates carotid ultrasound imaging and clinical data to assess stroke risk in hypertensive patients, addressing a critical need in contemporary clinical practice [21, 22]. To enhance the generalizability and accuracy of the model, a large-scale dataset was obtained from the publicly available Carotid Ultrasound Boundary Study (CUBS) database [23]. The CUBS dataset includes comprehensive ultrasound images, clinical parameters, and manually segmented carotid intima-media measurements. Specifically, the dataset comprises 2,176 ultrasound images from 1,088 patients, alongside clinical features such as age, sex, smoking history, and total cholesterol levels.

To ensure representative sampling and enhance model generalizability, a total of 1,088 hypertensive patients were enrolled and evenly stratified into a “high-risk group” ($n=544$) and a “low-risk group” ($n=544$) based on stroke risk assessment labels. The high-risk group included patients with a prior history of stroke or those clinically evaluated as extremely high-risk, while the low-risk group consisted of patients with no recent signs of stroke and normal clinical indicators. This balanced distribution was maintained throughout model training and testing.

In terms of demographics, the patient age ranged from 45 to 78 years, with a mean age of 61.3 ± 7.9 years. The cohort consisted of approximately 61.4% males and 38.6%

females. Among the cohort, 54.7% had a history of smoking and 28.5% had comorbid diabetes. All patients provided informed consent, underwent their first carotid ultrasound examination, and presented with complete clinical records and artifact-free imaging data.

Exclusion criteria were as follows: (1) presence of acute-phase cardiovascular or cerebrovascular complications; (2) low-quality ultrasound images or absence of key anatomical structures; (3) incomplete clinical records; and (4) lack of a recent hypertension diagnosis. These criteria ensured the quality and standardization of the dataset used for model development.

The dataset was curated and released by multiple collaborating institutions, including the Department of Electronics and Telecommunications at Politecnico di Torino, HELIOS Klinikum München West, and the University of Cyprus. All data used in the study comply with ethical guidelines and are available for research use under open-access provisions.

Display and preprocessing of carotid ultrasound image dataset

The carotid ultrasound image dataset included images with varying resolutions, which can negatively impact model training efficiency. Images with different resolutions require distinct computational resources and processing times, resulting in inconsistent processing speeds and a reduction in overall training performance. Furthermore, such variability introduced challenges in model generalization. To address these issues, all image data in this study were standardized to a resolution of 256×256 pixels during the preprocessing stage. For each patient, the dataset included ultrasound images acquired from both the left and right carotid arteries.

Comparative analysis of left and right carotid ultrasound images provides several important clinical insights: (1) Detection of asymmetry: Bilateral comparison enables the identification of differences in vessel size, blood flow velocity, and resistance [24], which may indicate potential abnormalities such as arterial stenosis, occlusion, or deformities. Although unilateral lesions may remain clinically silent, comparative analysis can uncover subtle pathological changes. (2) Determination of lesion extent and nature: Bilateral comparison helps establish whether lesions are confined to one carotid artery or are bilateral. For instance, unilateral carotid stenosis may indicate localized plaque formation, while bilateral stenosis may reflect underlying systemic vascular pathology. Comparative evaluation also facilitates differentiation between true lesions and anatomical variants. (3) Assessment of collateral circulation: In cases of stenosis or occlusion in one carotid or vertebral artery, compensatory expansion of collateral vessels on the contralateral side may occur to preserve cerebral perfusion. Bilateral assessment

supports the evaluation of collateral circulation development, which is critical for clinical decision-making. (4) Support for treatment decisions: Comparative results guide the selection of appropriate treatment options. For example, in the presence of severe unilateral stenosis with adequate contralateral perfusion, carotid endarterectomy or stenting may be indicated. (5) Monitoring of treatment outcomes: Following carotid endarterectomy or stenting, bilateral analysis helps monitor treatment efficacy and assess the occurrence of restenosis or other complications. (6) Evaluation of neck muscles and fascia: Ultrasound imaging can also evaluate the thickness and symmetry of neck muscles and fascia, contributing to the evaluation of muscular function and the identification of potential musculoskeletal issues [25].

To enhance image processing stability and ensure consistency during model training, all original carotid ultrasound images underwent standardized preprocessing steps: (1) Size normalization: Each image was resized to 256×256 pixels. (2) Grayscale intensity normalization: Pixel values were rescaled to the $[0, 1]$ range using min-max normalization. (3) Noise suppression: A Gaussian filter ($\sigma=1.2$) was applied to reduce image noise and eliminate background artifacts and blurred probe boundaries that might interfere with model learning.

For the annotation of the common carotid artery intima, all masks were independently labeled by two board-certified ultrasound specialists with over five years of clinical experience. A senior expert then reviewed and adjudicated the annotations. When the boundary difference between the two primary annotators exceeded 3 pixels, the expert-reviewed annotation was adopted as the final ground truth. Inter-observer agreement was assessed using the Dice Similarity Coefficient on a representative subset of the annotated dataset, yielding a mean score of 0.92, indicating high labeling consistency. This multi-expert annotation protocol ensured the accuracy and reproducibility of the training dataset.

In summary, bilateral comparison of carotid ultrasound images is crucial for a comprehensive evaluation of vascular health. This approach facilitates the detection of asymmetries and structural abnormalities, informs clinical decision-making, and supports the monitoring of treatment outcomes. Comparative analysis of both sides offers a more complete understanding of patient-specific vascular conditions, thereby improving individualized care strategies.

Clinical data presentation and preprocessing of carotid ultrasound dataset

To enhance diagnostic accuracy, treatment efficacy, and research capabilities, this study integrates clinical and imaging data to construct a high-performance stroke risk assessment model. Numerous studies have

demonstrated that integrating clinical and imaging data improves model performance, as illustrated by the following examples [26]: (1) Improved Diagnostic Accuracy: Imaging data provide detailed visualization of anatomical structures and pathological changes, while clinical data offer complementary information including patient history, presenting symptoms, physical findings, and laboratory results. The integration of these modalities allows for a more comprehensive evaluation of patient status, improving diagnostic accuracy. For instance, the detection of a pulmonary nodule on imaging becomes more clinically meaningful when interpreted alongside factors such as smoking history, cough, and hemoptysis, aiding in distinguishing between benign and malignant etiologies [27]. (2) Personalized Treatment: The integration of imaging and clinical data supports the development of individualized treatment strategies. For example, in patients with heart disease, imaging data offers detailed information about cardiac structure and function, while clinical parameters such as blood pressure, heart rate, and electrocardiographic data provide essential physiological context. This combination allows for accurate disease severity assessment and the selection of the most appropriate therapeutic interventions [28]. (3) Enhanced Healthcare Efficiency: Combined analysis of clinical and imaging data improves workflow efficiency by reducing redundancy in diagnostic procedures and minimizing unnecessary testing. Accelerated diagnostic processes contribute to timely clinical decision-making and optimized resource utilization.

To ensure data reliability and model robustness, quality control procedures were applied to the original clinical dataset. The curated and preprocessed clinical data are visualized in Fig. 1.

Intima-media segmentation model of the common carotid artery based on the ResNet network

To ensure optimal performance of the assessment model, this study employed ResNet, a widely used deep CNN. ResNet addresses key challenges in deep neural network training, including gradient vanishing and network degradation, through the introduction of residual blocks. These blocks enable the construction of deeper networks and improve overall learning performance [29, 30]. In this study, the ResNet50 model, which has a higher number of parameters, was chosen to further enhance segmentation accuracy, as shown in Fig. 2.

Although U-Net and its variants are widely adopted in medical image segmentation tasks, ResNet50 was adopted as the backbone feature extraction network for several key reasons. First, ResNet50's deeper convolutional architecture and residual connections allow for more effective extraction of edge information in low-contrast regions of carotid ultrasound images, thereby

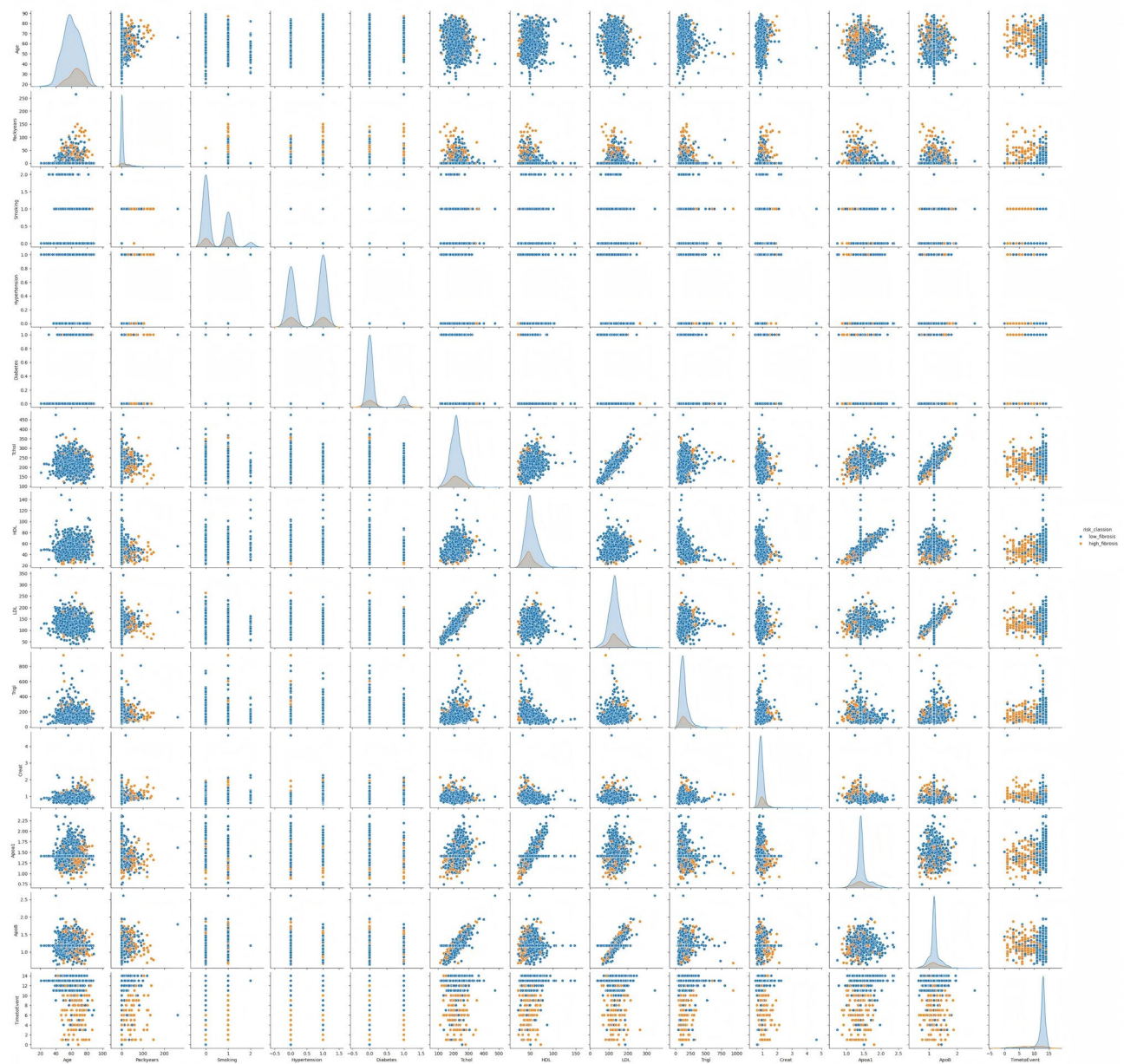


Fig. 1 Visualization of clinical feature distribution. Note: Blue indicates low-risk stroke patients, while yellow represents high-risk patients. Clinical features from left to right include: Age, Sex, CVD (Cerebrovascular Disease), Packyears (cumulative smoking exposure), Smoking status, Hypertension, Diabetes, Tchol (Total Cholesterol), HDL (High-Density Lipoprotein), LDL (Low-Density Lipoprotein), Trigl (Triglyceride level), Creat (Creatinine level), ApoA1 (Apolipoprotein A1 level), ApoB (Apolipoprotein B level), and TimetoEvent (frequency of cardiovascular events). Incorporating these multidimensional clinical features significantly enhances the performance of the stroke risk assessment model

improving the recognition of fine anatomical structures. Second, ResNet demonstrated strong generalization ability due to its pretraining on large-scale natural image datasets, which facilitated faster convergence in small-sample medical imaging tasks. Finally, the primary objective of the model involved classification based on global and regional receptive field features rather than pixel-wise semantic segmentation. The ability of ResNet50 to extract global representations made it particularly

suitable for integration with Transformer-based architectures. While the effectiveness of U-Net in semantic segmentation tasks is well recognized, future studies may consider comparative experiments between architectures to further evaluate performance differences.

The Bottleneck Block within ResNet50, a common module in CNNs, is essential for balancing computational efficiency with expressive capability. By compressing feature dimensions, the module reduces both computational

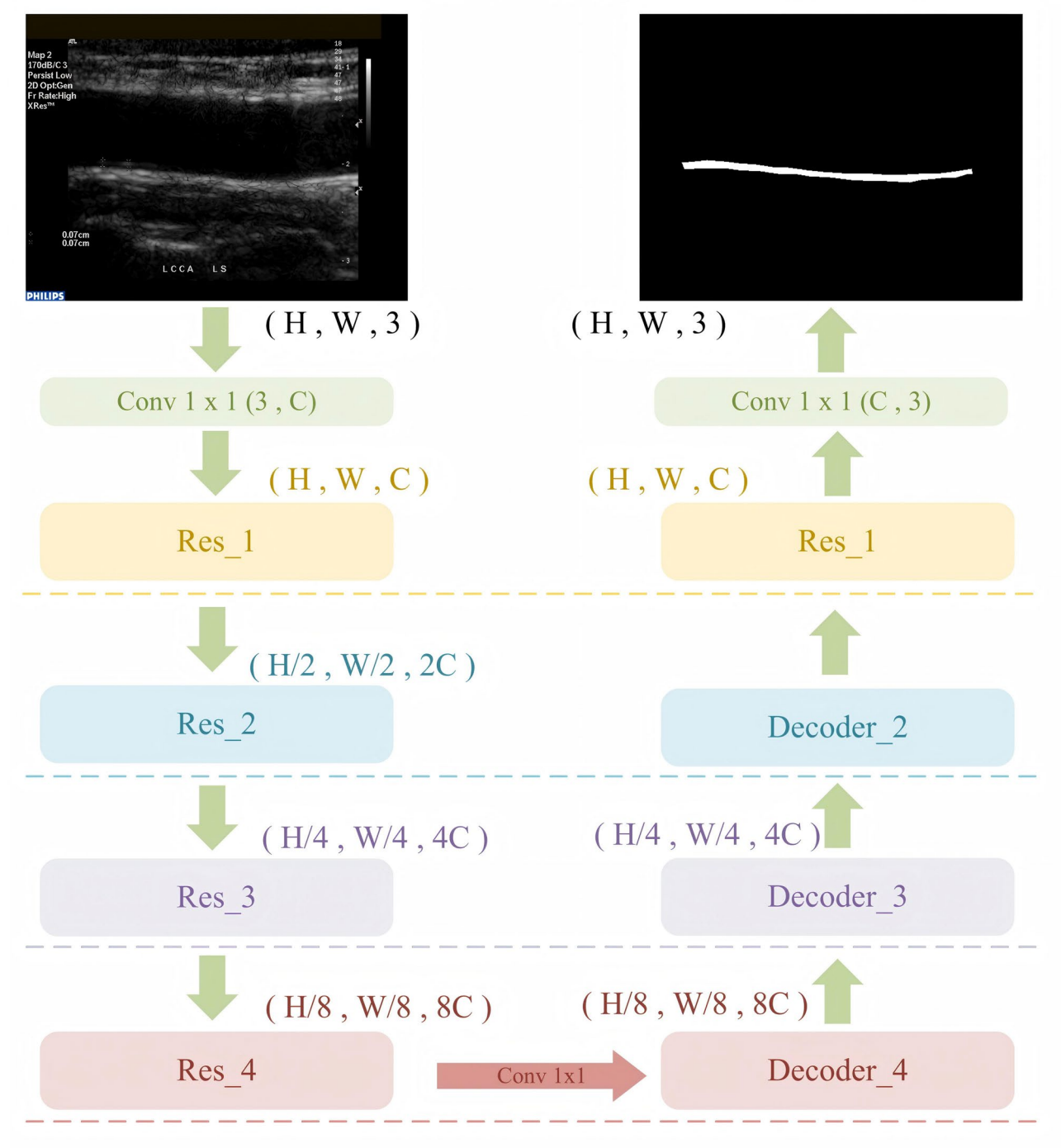


Fig. 2 Bottleneck block in the ResNet residual module. Note: The figure illustrates ResNet50 encoders on the left, each composed of multiple Bottleneck Blocks. From top to bottom, the four encoders (Res_1, Res_2, Res_3, Res_4) contain 3, 6, 4, and 3 Bottleneck Blocks, respectively, ensuring model performance while preventing overfitting

cost and parameter count while maintaining model performance, as illustrated in Fig. 3.

Several key advantages are associated with the Bottleneck Block: (1) Enhanced Segmentation Precision: Deeper networks with Bottleneck Blocks exhibit strong feature extraction capabilities, enabling the capture of

finer image details and improving segmentation accuracy. (2) Improved Feature Representation: Multi-scale feature information has been effectively integrated by applying the Bottleneck Block at various network layers. This approach captures richer contextual features and facilitates a more accurate interpretation of image

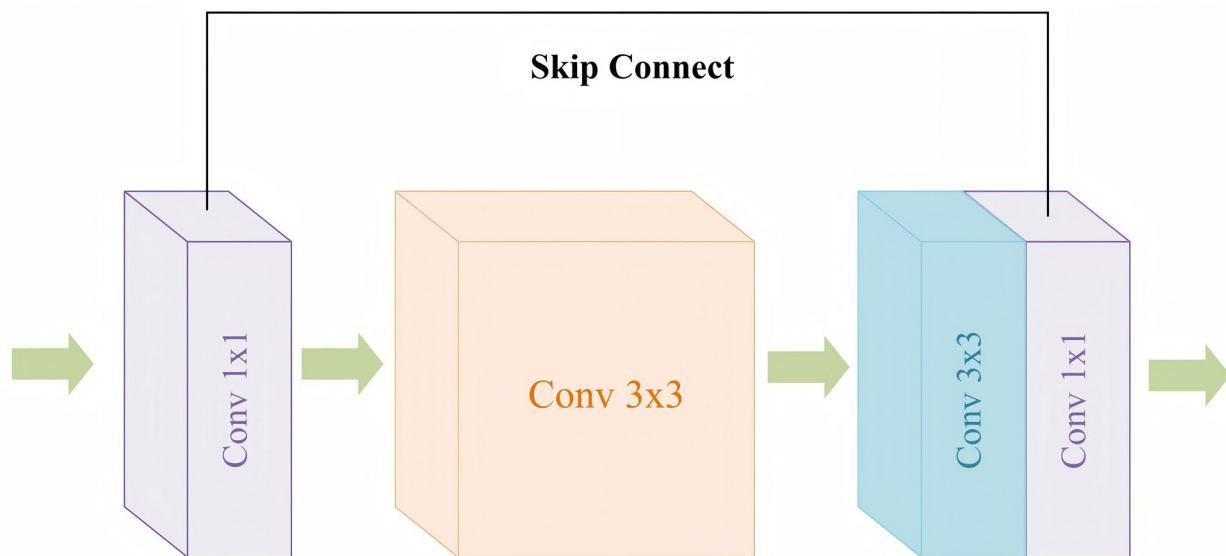


Fig. 3 Structure diagram of the bottleneck block. Note: The initial 1×1 convolution reduces the number of channels in the input feature map, effectively lowering the computational burden of the subsequent 3×3 convolution by reducing input dimensionality. The 3×3 convolution then extracts spatial features from the compressed representation, enabling effective capture of local spatial information. A final 1×1 convolution restores the original channel count, ensuring consistent feature dimensions and smooth propagation through subsequent layers in the network

content. (3) Higher Computational Efficiency: The reduction in computational complexity achieved by the Bottleneck Block has significantly improved the efficiency of the segmentation process, leading to faster inference. (4) Reduced Overfitting: The decrease in parameter count has contributed to lower overfitting risk and improved model generalization, especially when applied to datasets with limited sample sizes.

Stroke risk assessment model based on clinical feature integration

A Transformer-based architecture was adopted for stroke risk assessment, with the ViT serving as the core module due to its global receptive field advantage [31]. Unlike CNNs, ViT's self-attention mechanism captures relationships between any two pixels, effectively integrating global contextual information. This capability is crucial for fracture classification that depends on structural patterns extending beyond localized features. For instance, the accurate distinction between comminuted and simple fractures requires a comprehensive assessment of overall shape, damage extent, and surrounding tissue conditions. A global contextual view has been shown to improve model performance in such classification tasks. The primary Transformer module within ViT is illustrated in Fig. 4.

The ViT model used in this study was initialized with pretrained weights from ImageNet-21k, enabling accelerated convergence and enhanced feature representation during the early stages of training. Given the limited

sample size ($n=600$), transfer learning contributed to improved model stability. To further enhance feature clarity, especially in the presence of speckle noise commonly observed in carotid ultrasound imaging, a combined preprocessing strategy using a Gaussian filter ($\sigma=1.2$) and median filtering was applied. This strategy improved tissue boundary clarity and facilitated more robust feature extraction by the model.

As illustrated in Fig. 4, the model architecture from left to right begins with the ResNet50 down-sampling component, which extracts local details from ultrasound images, particularly the thickness of the carotid intima-media, a key indicator of stroke risk. An increase in carotid intima-media thickness often reflects the progression of atherosclerosis, a common precursor to stroke [32, 33]. Even in the absence of detectable carotid artery stenosis, increased intima-media thickness indicates an elevated risk of future stroke. The subsequent Attention module further extracts global information, enhancing the model's ability to assess intima-media thickness and improving its capability to identify high-risk individuals. After extracting imaging features, clinical data are flattened and integrated as linear features. These vectors are then concatenated with imaging-derived features to form a unified multidimensional representation, which is subsequently passed into the classification module for risk prediction.

The workflow of the proposed multimodal fusion model is structured as follows: each original carotid ultrasound image (size: 256×256) is first input into the

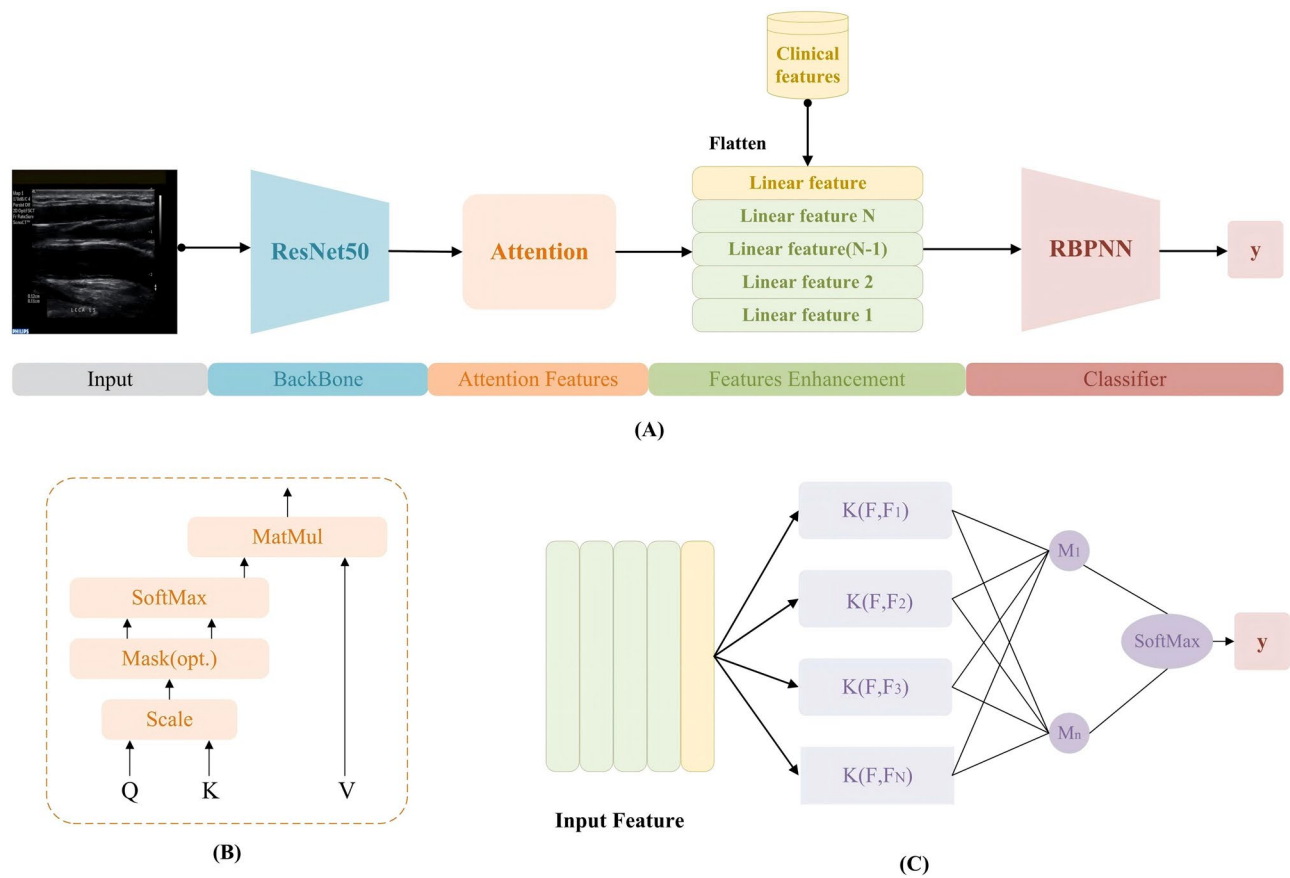


Fig. 4 Transformer module of the ViT model. Note: **(A)** shows the complete risk assessment model integrating carotid artery imaging data with clinical data. **(B)** details the self-attention computation process within the Transformer module. **(C)** visualizes the final classification output using the RBPNN

ResNet50 network. Deep semantic features are extracted via residual blocks (Res_1 to Res_4), followed by a Global Average Pooling layer that generates a fixed-length feature vector of size 1×2048 . This vector is then passed to the Vision Transformer (ViT) module, where it is treated as a visual token embedded from image patches. Combined with standard positional encoding, the token is input into a multi-head self-attention mechanism to learn global dependencies among image features, resulting in a structured semantic representation vector of the image.

Simultaneously, clinical features (14 variables) are standardized using Z-score normalization and imputed for missing values during preprocessing to form a 1×14 vector. This vector is flattened and concatenated with the ViT-derived image feature vector, producing a final fused feature vector of size $1 \times (2048 + 14)$.

This multimodal feature vector is then fed into the Radial Basis Probabilistic Neural Network (RBPNN) classifier. The first two layers—Radial Basis Function transformation and probabilistic mapping—enable stroke risk prediction for each sample, producing a probability score indicating classification into the “high-risk” or “low-risk” group.

The classification model, Radial Basis Probabilistic Neural Network (RBPNN), combines the strengths of Radial Basis Function (RBF) networks and Probabilistic Neural Networks (PNNs). RBPNN is an effective classifier for nonlinear data and demonstrates strong generalization performance [34], as shown in Fig. 4(C). The architecture consists of three main layers. The first layer, the input feature layer, receives both imaging-derived features and clinical variables, including age, gender, total cholesterol, blood pressure, and blood glucose levels. The second layer, the radial basis layer, comprises multiple RBF units, each associated with a prototype vector. Each RBF unit calculates the distance between the input vector and its prototype vector, outputting a value (typically a Gaussian function) representing the similarity between the input vector and the prototype. The final layer, the output layer, calculates the probability distribution across classification categories. The outputs from the radial basis layer are weighted, summed, and normalized using a softmax function to generate class probabilities. The class with the highest probability is selected as the final prediction. The output mechanism aligns with the structure of PNNs, enhancing the interpretability and

robustness of the model in clinical classification tasks [35].

The clinical variables used in this study included age, sex, systolic blood pressure, diastolic blood pressure, BMI, smoking status, comorbid diabetes, total cholesterol, fasting blood glucose, and heart rate, totaling 10 features. For variables with <10% missing data, mean imputation was applied. Features with >20% missingness were excluded from training. After preprocessing, nine clinical features were retained and standardized via Z-score normalization.

A mid-level fusion strategy was adopted to integrate features: image representations extracted from ResNet50 and ViT (dimension=512) were concatenated with clinical features processed through a fully connected network (dimension=32). The combined vector was then passed through a multilayer perceptron (MLP) for further dimensionality reduction and integration. This fusion approach enabled joint modeling of structural and phenotypic information. Although an attention-based fusion mechanism was initially tested, no significant performance improvement was observed during preliminary testing. Therefore, a simpler concatenation-based fusion was selected to preserve architectural clarity and ensure stable training.

Model training and validation strategy

To ensure robust and reliable model evaluation, strict controls were applied during dataset partitioning and validation. All samples were stratified and randomly split by patient ID into training, validation, and test sets, thereby preventing data leakage by ensuring that multiple images from the same patient did not appear in different subsets.

A five-fold cross-validation scheme was used to assess model robustness. Early stopping (patience=10), dropout regularization ($p=0.4$), and L2 regularization were employed to prevent overfitting. To address class imbalance, a weighted loss function and data augmentation techniques were implemented to enhance the model's ability to recognize minority class samples.

Model training was performed using the Adam optimizer with an initial learning rate of $1e-4$. A learning rate decay schedule was applied, reducing the rate by a factor of 0.1 every 10 epochs. The batch size was set to 16, and the maximum number of training epochs was limited to 100. Cross-Entropy Loss served as the objective function. All training and inference were conducted on an NVIDIA RTX 3080 GPU. Early stopping was triggered when the validation AUC failed to improve over 10 consecutive epochs.

All deep learning models were initialized with ImageNet-pretrained weights to enhance feature extraction and accelerate convergence. The ViT module was configured with a patch size of 16, an embedding dimension of

768, 12 transformer encoder layers, each with 12 attention heads, an MLP hidden size of 3072, and a dropout rate of 0.1. Full architectural parameters and training configurations are detailed in Supplementary Table 1 to ensure reproducibility in future studies.

Statistical analysis

A range of software tools and statistical methods were employed to develop and evaluate a predictive model for surgical outcomes and postoperative recovery. All programming and data preprocessing was conducted in Jupyter Notebook, using Python as the coding language. Data visualizations were performed using Visio, while data manipulation and plotting were carried out with Python's pandas and Matplotlib libraries. Statistical procedures included data normalization, accuracy calculation, sensitivity and specificity analysis, and evaluation of model performance using loss function curves.

During the data processing phase, image data were normalized to eliminate scale differences among features. Additionally, all images were also resized to a uniform resolution to ensure consistent input for model training.

In the evaluation phase, accuracy, sensitivity, and specificity were calculated to assess model performance. Accuracy measured the overall predictive capacity, sensitivity reflected the ability to correctly identify positive cases, and specificity quantified the ability to detect true negatives. Together, these metrics provided a comprehensive assessment of the model's clinical applicability.

Loss function curves were analyzed to further evaluate training quality and convergence behavior. Inspection of curve trends and divergence between training and validation loss enabled the identification of potential overfitting and facilitated model optimization.

To visualize results, Matplotlib was used to generate plots that illustrated spatial feature distributions and model performance metrics. This visualization supported the interpretation of data characteristics and provided insight into model behavior under various conditions. The combined application of statistical analysis and visual tools allowed for thorough validation of model reliability and applicability in real-world clinical scenarios.

Results

Qualitative analysis of the carotid artery intima segmentation model based on Resnet

This study employed a ResNet segmentation network to precisely segment the carotid artery intima, enabling accurate measurement of intima thickness—a critical metric for predicting stroke risk in hypertensive patients. Qualitative analysis indicated that the model effectively captures the carotid intima boundaries across different patients, maintaining high segmentation accuracy even in complex or low-contrast ultrasound images. As

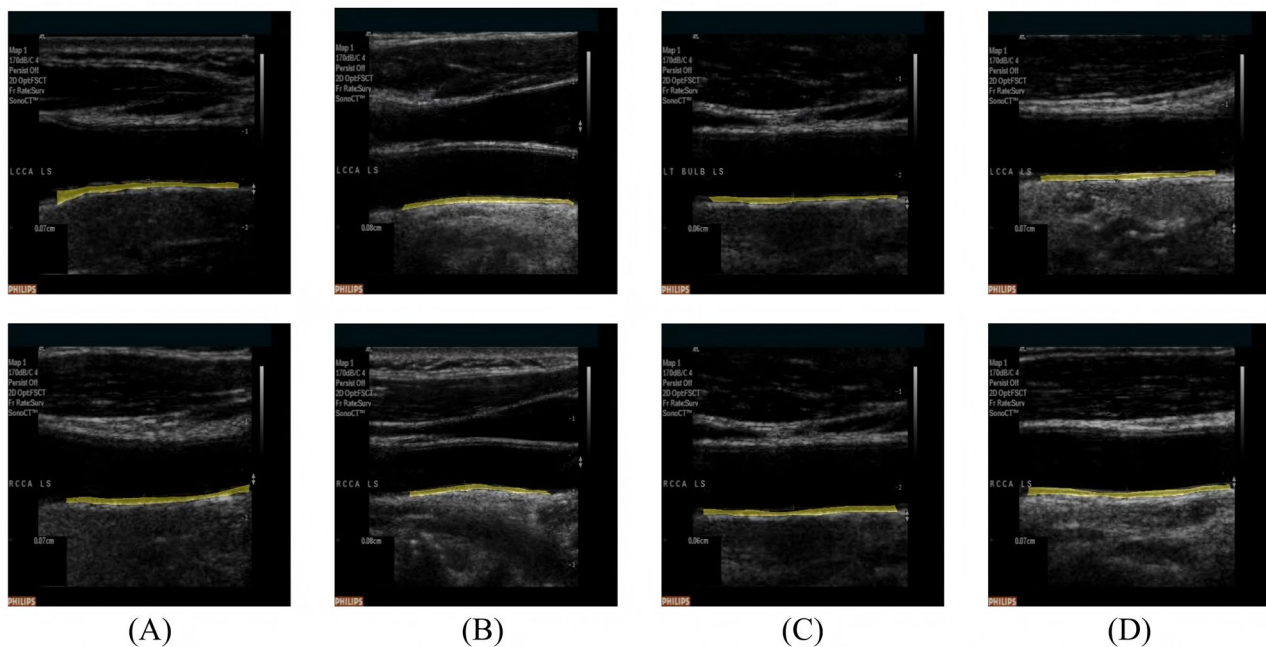


Fig. 5 Carotid artery intima segmentation results. Note: (A), (B), (C), and (D) represent ultrasound images from four randomly selected patients. The first and second columns display the left and right carotid arteries, respectively. Intima thickness features extracted from both sides contribute to the model's subsequent risk prediction performance

illustrated in Fig. 5, bilateral analysis of intima thickness enhanced the model's ability to identify potential arterial lesions and assess stroke risk. Throughout the training process, both the Dice coefficient and Intersection over Union (IoU) metrics showed progressive improvement, validating the effectiveness of the ResNet50's Bottleneck Block in complex medical image segmentation. This module improved the capture of boundary details while significantly reducing the model's parameter count and computational complexity, boosting both training and inference efficiency.

In clinical applications, measurement of carotid intima thickness on both sides enables comparison of bilateral differences, offering valuable insights for early identification of stroke risk in hypertensive patients. Clinical studies have demonstrated a strong association between increased intima thickness and the progression of atherosclerosis. Among asymptomatic patients, significant disparities in bilateral intima thickness often indicate potential underlying vascular abnormalities. The qualitative analysis of the segmentation results confirmed that the ResNet segmentation network accurately delineated carotid intima boundaries, even in areas with blurred vascular edges, thus providing a reliable anatomical foundation for subsequent lesion detection.

Qualitative analysis of the carotid intima segmentation model based on Resnet segmentation network

This study conducted a qualitative analysis of the carotid artery segmentation model by visualizing changes in key training metrics, as shown in Fig. 6. The ResNet50 segmentation model demonstrated strong convergence and robustness, with a consistent decline in the training loss curve and stable Dice and IoU coefficients on the test set. These results suggest that the model can effectively generalize to previously unseen data. Notably, accurate segmentation of the carotid intima was achieved even in ultrasound images with low contrast and irregular boundaries, reducing uncertainty and errors associated with manual annotation. The loss function curve and IoU trend shown in Fig. 6 further validated the model's stability when processing unstructured ultrasound data. This segmentation performance can be attributed to ResNet's unique residual connections (Fig. 2, Bottleneck Block), which enabled direct transmission of input information to subsequent layers. This structural design improved the network's ability to capture essential texture and boundary details while reducing the risk of gradient vanishing during training.

As shown in Fig. 6A, the training loss stabilized around 0.5 after 10 epochs, indicating high training stability and strong generalization capacity.

Figure 6B shows the Dice Similarity Coefficient (DSC), a commonly used metric for evaluating segmentation performance by measuring the overlap between

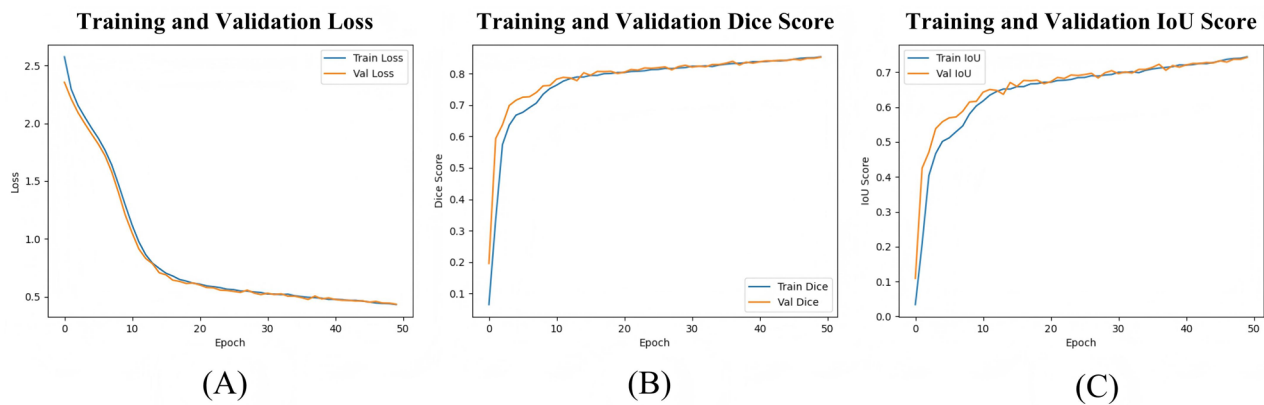


Fig. 6 Visualization of feature extraction across layers in the prognostic model. Note: (A), (B), and (C) represent training curves for the loss function, Dice similarity coefficient, and intersection over union (IoU), respectively, reflecting model convergence and segmentation accuracy

predicted and actual segmented regions. The DSC ranges from 0 to 1, with values closer to 1 indicating better alignment. In this study, the model achieved a DSC of approximately 0.9, demonstrating high segmentation accuracy in capturing the carotid artery intima boundary. A DSC close to 0.9 reflects approximately 90% overlap between the predicted segmentation and the reference annotation, providing a reliable basis for subsequent measurements and analysis. The consistently high DSC further confirmed the robustness, generalizability, and high segmentation accuracy of our model.

As shown in Fig. 6C, the IoU metric, commonly used in image segmentation tasks, was employed to evaluate the overlap between the model's predicted segmentation and the true segmentation. IoU values range from 0 to 1, where 1 indicates perfect overlap, and 0 indicates no overlap. For medical image segmentation, IoU serves as a key indicator of the model's ability to delineate anatomical boundaries (e.g., carotid artery intima). Higher IoU values indicate improved accuracy in delineating object shape and position, making this metric particularly suitable for applications requiring precise boundary identification. In this study, the ResNet segmentation model achieved an IoU of approximately 0.8 on the carotid artery intima segmentation task, indicating high accuracy and stability in segmenting this region. An IoU of 0.8 suggests that the predicted segmentation closely approximated the ground truth, with reduced error compared to manual annotation or models with lower IoU. Medical image segmentation often requires capturing fine details beyond macrostructures. Achieving an IoU of 0.8 signifies that the model aligned with clinical expectations at a macro level while accurately identifying local irregularities, such as local thickening or blurred edges. From a performance standpoint, an IoU of 0.8 further underscores the model's robustness in handling various complexities. The boundary of the carotid artery intima can

appear blurred due to image quality, noise, and contrast variations. Despite these challenges, the model maintained consistently high IoU across diverse data, demonstrating strong robustness to input variability. This stability improved reproducibility and ensured reliability across different patients, devices, and conditions, thereby enhancing clinical adaptability. Finally, from a clinical perspective, an IoU of 0.8 provided a reliable basis for downstream stroke risk stratification. Carotid intima thickness is closely related to stroke, hypertension, and other conditions. Accurate segmentation supports clinicians in quantifying intima thickness more precisely. The consistently high IoU not only validated the model's technical performance but also reflected its potential value in clinical applications, such as stroke risk assessment and hypertension management.

Quantitative analysis of a stroke risk assessment model integrating clinical features

Ultrasound imaging features and multidimensional clinical data were integrated to develop a stroke risk assessment model aimed at improving the early identification of high-risk stroke patients. Compared to traditional risk assessment methods, the proposed model leveraged a broader set of features, enabling more accurate and stable risk prediction.

Model performance was quantitatively evaluated using the Precision-Recall curve and the Receiver Operating Characteristic (ROC) curve, as shown in Fig. 7. The model achieved an Area Under the Curve (AUC) of 0.97, demonstrating high accuracy in distinguishing between high- and low-risk patients. Strong performance was also observed in terms of precision and recall, indicating robust sensitivity and specificity under practical conditions. Additionally, the quantitative analysis demonstrated that the model, incorporating multiple clinical variables (e.g., age, gender, total cholesterol, smoking

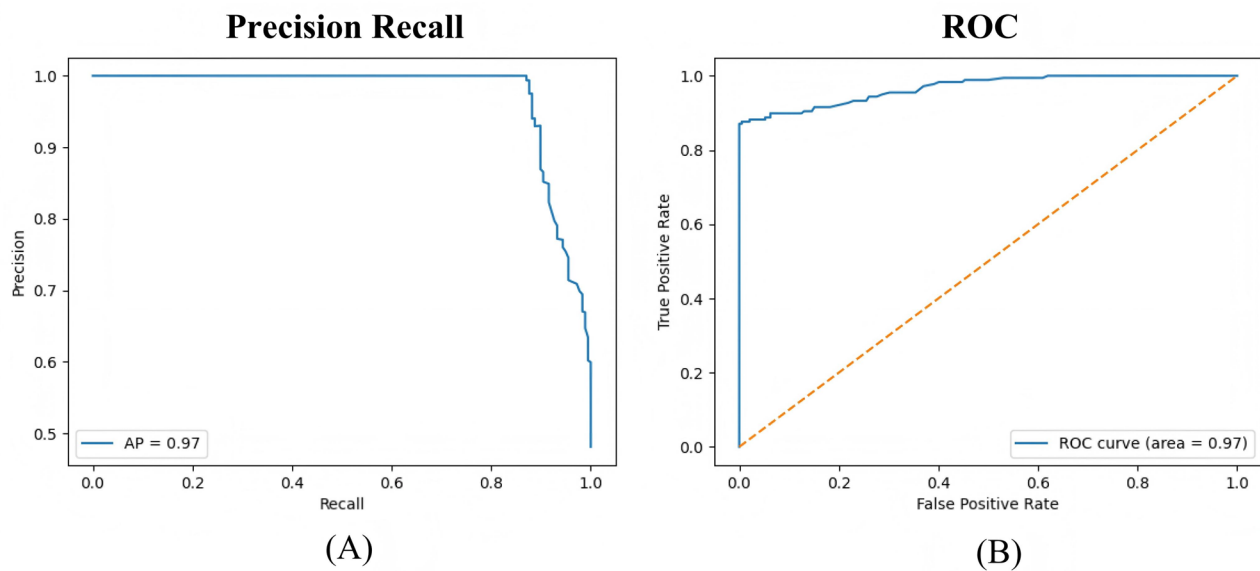


Fig. 7 Visualization of feature extraction across layers in the prognostic model. Note: (A) and (B) display the Precision-Recall and ROC curves for the risk assessment model, respectively

history, hypertension, and diabetes), significantly outperformed models based solely on imaging or a single variable. These results highlighted the value of multimodal data fusion in improving predictive performance and emphasized the importance of comprehensive clinical integration in stroke risk modeling.

In terms of imaging feature extraction, the model initially employed ResNet50 to process ultrasound images, capturing critical anatomical information such as carotid intima-media thickness, a significant marker for stroke risk assessment. The Transformer-based architecture subsequently integrated these imaging features with multidimensional clinical data, enabling adaptive multi-layer feature extraction and analysis. By leveraging the Transformer's global self-attention mechanism, the model effectively extracted global structural patterns within the images while simultaneously capturing interdependencies among clinical variables, thereby enhancing stroke risk quantification accuracy. To evaluate generalizability, the model was tested across multiple independent datasets. The results demonstrated consistently high accuracy across diverse data centers, underscoring its potential for widespread clinical application. Notably, the model demonstrated robust predictive consistency across stratified populations, especially among patients with combined high-risk factors like hypertension and diabetes, where sensitivity and specificity are significantly enhanced to meet clinical standards.

In summary, the stroke risk assessment model integrating clinical and imaging features demonstrated superior quantitative performance and substantial clinical applicability. The model provided an effective tool for

personalized prevention and management of stroke risk in hypertensive patients. With ongoing refinement of clinical variables and feature fusion techniques, further advancements in the precision and intelligence of stroke risk prediction are expected.

Model comparison and ablation study

To further evaluate the contribution of each module within the proposed model architecture, an ablation study was conducted comparing three configurations for high stroke risk prediction: (1) ResNet50 for image feature extraction+fully connected classifier; (2) ResNet50+Vision Transformer (ViT) for enhanced image feature extraction; (3) Full model: ResNet50+ViT+RBPN (referred to as the proposed method).

The results demonstrated that the full model outperformed the simplified variants across multiple metrics, including AUC, precision, recall, and IoU (Supplementary Table 2), indicating the significant value of deep multimodal fusion in enhancing predictive performance.

Additionally, subgroup analysis was performed on high-risk populations, including hypertensive patients with comorbid diabetes, smokers, and non-smokers. The model exhibited robust performance across these subgroups, supporting its potential applicability in real-world clinical risk stratification.

To comprehensively assess the effectiveness of the proposed fusion model, we introduced two benchmark comparators: (1) the Framingham Stroke Risk Score, a classical clinical risk scoring system based on demographic and clinical parameters such as age, blood

pressure, smoking status, and diabetes; (2) DenseNet, a widely used deep learning architecture that relies solely on ultrasound images, representing an image-driven end-to-end learning approach.

All models were evaluated on the same test dataset using five key metrics: AUC, accuracy, precision, recall, and F1 score. As shown in Supplementary Table 3, the proposed ResNet50 + ViT + Clinical fusion model outperformed both baselines across all five metrics, with particularly strong results in AUC (0.920) and F1 score (0.885), indicating its strong capability in identifying high-risk stroke patients with greater accuracy and reliability.

Discussion

A deep learning-based multimodal integration framework is employed to automate carotid artery ultrasound image segmentation and predict stroke risk in hypertensive patients, offering both theoretical innovation and clinical relevance.

Firstly, the proposed multimodal data fusion model effectively combines imaging data with multidimensional clinical features, offering a novel technical perspective for stroke risk assessment. Unlike traditional methods that process medical imaging and clinical data separately, this study utilizes an RBPNN mechanism to enable comprehensive feature interaction across both data types, thereby improving predictive accuracy in complex clinical scenarios. Integration of deep learning with multimodal data fusion enables a more comprehensive approach to stroke risk prediction in hypertensive patients [36]. Compared to traditional single-source risk assessment methods, this fusion model captures the nonlinear relationships among diverse features more effectively, addressing key limitations in adaptability and representational capacity. By facilitating hierarchical integration of clinical and imaging information, the model enhances predictive performance across multiple dimensions, including accuracy, sensitivity, and specificity. The demonstrated robustness and generalizability support its clinical applicability and potential for large-scale deployment.

Additionally, the ResNet50 deep convolutional network used in this study demonstrated high accuracy and robustness in the automatic segmentation of carotid artery intima, enhancing the reliability of the stroke risk assessment model [37]. Accurate measurement of intima thickness provides a solid data foundation for quantifying pre-stroke pathological features, such as atherosclerosis, and facilitates the early identification of high-risk individuals during asymptomatic phases. The segmentation framework reduces the manual burden on clinicians and enables non-invasive, real-time stroke screening methods. Furthermore, this non-invasive ultrasound-based assessment is suitable for a broader population of

hypertensive patients, with potential for use in community screenings and home health management.

Model evaluation across multicenter datasets with varied ultrasound imaging conditions and clinical profiles confirms its adaptability and generalizability. Multicenter experiments confirmed consistent performance across different ultrasound systems and demographic profiles, highlighting its applicability in diverse clinical settings including hospitals, specialty centers, and remote healthcare environments. In an era of increasing artificial intelligence applications, the model's automated and personalized features align with the advancements in precision medicine and smart healthcare, offering new insights for improving clinical diagnostic efficiency and optimizing healthcare resource allocation.

To further evaluate the real-world applicability of the proposed model, we analyzed its computational complexity. The complete model (ResNet50 + ViT + RBPNN) demonstrated an average inference time of 0.84 s per image, with approximately 34.9 million parameters and 5.2 GFLOPs. Compared with the baseline ResNet50 + fully connected model (23.5 M parameters, 3.4 GFLOPs), the increase in computational load remains moderate. The model still supports real-time inference, making it suitable for high-throughput screening and individualized assessment. Future optimization strategies such as model pruning or knowledge distillation may further enable deployment on resource-constrained devices.

Compared with previous studies on stroke risk prediction, this research demonstrates significant advantages through the integration of multimodal data fusion and DL-based modeling. Many previous studies rely on either isolated imaging data or a limited set of clinical features for risk assessment, resulting in reduced model robustness and high sensitivity to data dimensionality and quality constraints [38, 39]. For instance, traditional logistic regression models generally consider only basic patient history or imaging data, lacking the capability to integrate multidimensional clinical and imaging information, which limits predictive accuracy [8, 40, 41]. Although convolutional neural networks such as U-Net have contributed to advances in medical image segmentation, these models still exhibit limited precision and poor generalizability across heterogeneous populations [39]. The combination of the RBPNN mechanism and ResNet50 enables more accurate and robust modeling of nonlinear relationships between imaging and clinical features, providing a more comprehensive and precise reference for stroke risk assessment. These improvements effectively address the limitations encountered in previous single-modality studies. Multicenter data validation further enhances the model's clinical applicability and generalizability.

Although this study demonstrates promising results, several limitations remain. First, the limited sample size and lack of diversity may restrict the model's applicability to specific groups, including elderly individuals, patients from varied ethnic backgrounds, and those with multiple comorbidities. Further validation in a broader population is necessary to confirm model stability and generalizability. Second, the performance of the ultrasound image segmentation technique may be affected by poor image quality or small lesion size, highlighting the need for further refinement of preprocessing and enhancement techniques. Additionally, the "black-box" nature of DL models limits their interpretability. Future research should prioritize the development of more transparent architectures or post hoc interpretability tools to improve clinical trust. Finally, this study primarily focused on hypertensive patients, without extending to other high-risk groups such as individuals with diabetes or coronary artery disease. Additional research is required to evaluate the model's performance across these broader patient cohorts.

Although the CUBS dataset used in this study includes multicenter carotid ultrasound images and corresponding clinical data, which enhances model robustness and generalizability, several limitations remain. The dataset is primarily sourced from specific geographic regions and institutions, and the patient population may lack diversity in age, ethnicity, and disease spectrum, potentially leading to performance fluctuations in broader populations. Additionally, certain important stroke-related risk factors, such as lifestyle or genetic data, are not included. The current model also focuses solely on ultrasound data without integrating other imaging modalities (e.g., CT or MRI), limiting its applicability in multi-modal diagnostic contexts. Future studies should consider incorporating larger, multi-regional, and multi-disease datasets to improve external validity.

While the model achieves high accuracy in identifying high-risk stroke patients, accuracy alone does not ensure clinical applicability. Based on literature review and expert consultation, a risk threshold of 0.7 is initially established for clinical intervention (i.e., individuals with model output > 0.7 are recommended for further evaluation or follow-up). Sensitivity and specificity analyses across thresholds revealed that 0.7 offers a favorable balance between recall and false-positive control. The clinical implications of misclassification are also considered. False-positive predictions may lead to unnecessary consumption of medical resources or increased patient anxiety, whereas false-negative results may delay timely interventions. Future validation on real-world cohorts is planned to assess clinical outcomes under different threshold settings.

The average inference time per case was approximately 45 ms when executed on an NVIDIA RTX 3080 graphics processing unit, enabling real-time clinical deployment. On a moderately configured CPU (Intel i7, 32GB RAM), the inference time increased to around 120 ms per case, which remains acceptable for routine use in hospital environments. Nonetheless, several challenges remain for clinical deployment: (i) system heterogeneity across hospital infrastructures requires interface adaptation; (ii) regulatory constraints on data privacy limit the feasibility of cloud-based deployment; and (iii) lower-tier medical institutions may face constraints in computing infrastructure. Future development will prioritize the design of lightweight model design, edge deployment, and standardized interfaces to improve clinical usability.

While this study primarily focused on improving prediction performance, fairness represents a critical dimension in clinical model development. Due to the lack of race-related data in the public dataset, population-level fairness assessments based on race could not be performed. A preliminary subgroup analysis based on sex revealed minimal bias, with AUCs of 0.918 for males and 0.923 for females. However, fairness extends beyond statistical parity and encompasses equitable access to medical resources. In the future, we plan to incorporate more diverse population datasets and adopt adversarial de-biasing strategies to enhance group-level robustness and fairness. To mitigate the risk of unintended disparities, model validation on local population structures is strongly recommended prior to clinical implementation. Future work will focus on the following aspects: (1) External validation using larger, multi-center datasets with more demographically diverse populations; (2) Integration of explainable AI techniques (e.g., SHAP analysis, counterfactual reasoning) to improve clinical transparency and acceptance; (3) Application of model compression and lightweight architecture design to support real-time inference and deployment in edge devices or hospital information systems.

Conclusions

In summary, the proposed stroke risk assessment model integrates DL and multimodal data fusion to achieve efficient integration of medical imaging and clinical data. The approach demonstrates strong potential for clinical implementation due to its scalability and predictive accuracy. With further data enrichment and model optimization, this approach holds potential for broader use in early screening, personalized intervention, and management of high-risk stroke populations.

Abbreviations

ApoA1	Apolipoprotein A1
ApoB	Apolipoprotein B
AUC	Area Under the Curve

CNN	Convolutional Neural Network
Creat	Creatinine
CUBS	Carotid Ultrasound Boundary Study
CVD	Cerebrovascular Disease
DL	Deep Learning
DSC	Dice Similarity Coefficient
HDL	High-Density Lipoprotein
IoU	Intersection over Union
LDL	Low-Density Lipoprotein
PNN	Probabilistic Neural Network
RBF	Radial Basis Function
RBPNN	Radial Basis Probabilistic Neural Network
ResNet	Residual Neural Network
ROC	Receiver Operating Characteristic
Trigl	Triglyceride
ViT	Vision Transformer

Supplementary Information

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Supplementary Material 1

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Author contributions

Hui Li and Tianyu Zhang (co-first authors): Conceptualization, Methodology, Data Curation, Writing – Original Draft. Guochao Han and Zonghui Huang: Investigation, Formal Analysis, Software. Huiyu Xiao and Yunzhe Ni: Visualization, Validation, Data Processing. Bo Liu and Wennan Lin: Clinical Data Collection, Resources, Supervision. Yuan Lin (Corresponding Author): Project Administration, Funding Acquisition, Writing – Review & Editing, Final Approval. All authors critically reviewed and approved the final manuscript.

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Data availability

The datasets used or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Relevant guidelines and regulations

Not applicable.

Declaration of the use of generative AI and AI-assisted technologies in the writing process

We did not use the generative AI and AI-assisted technologies to write the manuscript. Generative AI and AI-assisted technologies were only used to check grammar and spelling errors after writing the manuscript. After using this service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Competing interests

The authors declare no competing interests.

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